EVIDENCE-FIRST PORTFOLIO

ECSA predictive AMR AI architecture

Individual printable topic report.

Generated 2026-06-17

Why ECSA belongs here

ECSA is my future project answer: an AI-for-science direction where uncertainty, governance, validation, and model accountability matter. It should not replace the Secure AI infrastructure proof, but it gives the Fellowship a concrete research direction.

The architecture models antimicrobial resistance as a probabilistic graph/state-space problem with resistance, uncertainty, and stability rather than one static black-box prediction.

- Start with synthetic/public data before clinical data.
- Build Bayesian update behavior and graph learning under explicit uncertainty.
- Add governance, audit logs, model cards, and intended-use boundaries before real-world claims.
- Use lab-in-the-loop validation only after a reproducible proof of concept exists.

POC sequence

Stage	Build	Success criterion
1. Synthetic/public simulator	Small AMR state-space simulator with R, U/sigma, S, transitions, and noisy observations.	Reproducible behavior without private clinical data.
2. Bayesian update layer	Uncertainty-aware updates from synthetic or literature-derived observations.	Calibration and uncertainty shrinkage can be measured.
3. Graph learning layer	Link prediction or graph neural network experiments for missing collateral-sensitivity relationships.	Predictions compared against withheld or simulated links.
4. Governance/audit layer	Model cards, provenance, intended-use boundary, failure modes, and human review gates.	No output can be mistaken for validated clinical advice.
5. External validation plan	Lab/stakeholder protocol before real-data claims.	Independent expert review gates the next step.

Connection to my agent systems

ECSA is not random next to Miguel. The same engineering concerns transfer: model routing, evaluator loops, memory provenance, uncertainty, auditability, and human-in-the-loop control.

A Miguel-like infrastructure could support literature ingestion, hypothesis generation, experiment tracking, model evaluation, and governance documentation, while ECSA gives that infrastructure a high-stakes scientific target.